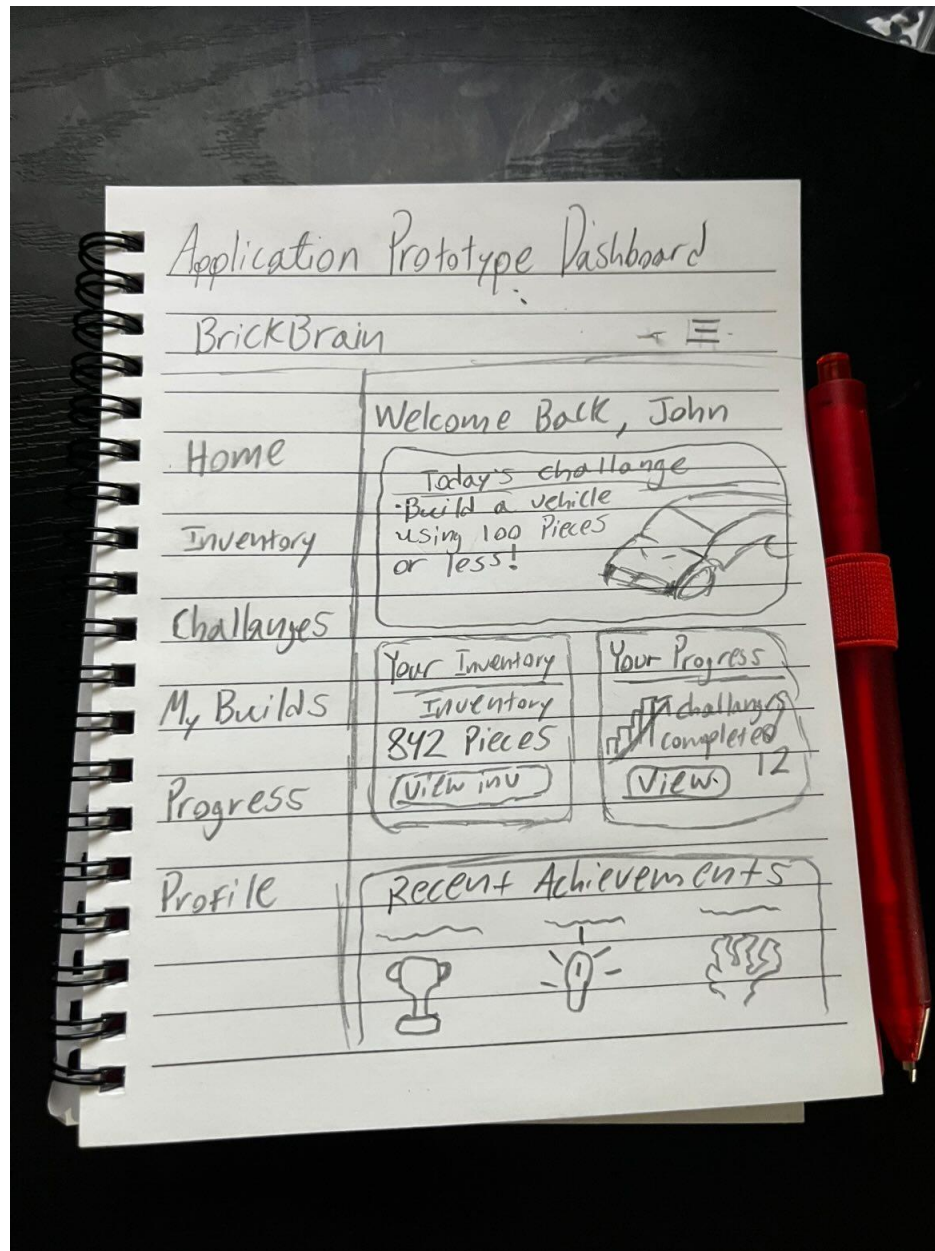


BrickBrain: Intelligent LEGO Build System

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Technical Field

This project involves the technical fields of Artificial Intelligence, Software Engineering, Database Management, Mobile Application Development, and Computer Vision.

Background Information

LEGOs and building them is a very big hobby for many people all over the world. A hobby that promotes creativity, spatial reasoning, and engineering-based problem solving as well. Many LEGO enthusiasts tend to gather lots of LEGO sets and proceed to build a large collection. Some of these collections are often underutilized because a majority of the time people find it difficult to identify new build possibilities with a lot of left over and extra parts.

Many current LEGO tools focus too much on providing official set instructions or trying to provide other designs with digital environments or fields. Even though these are useful, they aren't as effective in helping users discover new ideas and challenges to open the creative door towards builds based off their inventory.

This project was inspired by my passion for LEGOs and wanting to help other enthusiasts like myself to help others interact with their existing collections and create challenges for themselves to improve their skill in problem solving and creativity. The goal is to use artificial intelligence to transform this into a dynamic and more creative opportunity for users while also encouraging an innovative and experimentation in design.

Prior Art (Research)

Prior Art

Brickit

- Brickit is a mobile application that helps LEGO users identify pieces from their collection and provides suggestions for models they can build using those available parts. The platform focuses on inventory analysis and matching existing designs to a user's collection.

How BrickBrain Differs

- BrickBrain differs from Brickit because it doesn't focus on recommending existing models. Instead, BrickBrain functions as an adaptive LEGO design coach system that generates personalized build challenges based on a user's inventory, interests, and skill level. The goal is to push original creations and skill development rather than just simply matching pieces a user has to pre-existing designs.

Citation

Brickit. (2025). Brickit App. <https://brickit.app>

Rebrickable

- Rebrickable is an online LEGO inventory and database application that allows users to track their collections, discover existing builds, and help identify missing pieces required for specific builds. ReBrickable focuses on inventory managing and finding MOCs or official sets.

How BrickBrain Differs

- BrickBrain differs from Rebrickable because its primary purpose isn't inventory tracking or finding existing builds. Instead, BrickBrain creates adaptive building challenges that help encourage users to create original LEGO designs using their available pieces on a build. By combining the options of themes, difficulty levels, and challenge requirements, BrickBrain promotes creativity, experimentation, and skill growth rather than simply providing model recommendations.

Citation

Rebrickable. (2025). Rebrickable LEGO Database. <https://rebrickable.com>

Project Description

BrickBrain is a LEGO design coach system that generates build challenges that are based on a user's inventory, interests, and skill level. Users then have the option to choose the difficulty based off their experience, rather than simply recommending existing LEGO models, the system encourages users to create original builds by providing creative constraints, themes, and objectives solely based off the pieces they have in their system.

BrickBrain combines inventory management, AI, and challenge generation to transform a LEGO collection into interactive learning and creativity experiences that would help a user's skill development. By analyzing a user's available pieces and preferences, BrickBrain will create building activities and challenges that'll create problem-solving, experimentation, and design thinking.

This project targets LEGO hobbyists, collectors, students, and creative builders who want to improve their building skills while making greater use of their existing collections. BrickBrain achieves this by providing personalized challenges and guided creativity, BrickBrain helps users discover new ways to build and learn rather than simply following instructions from existing sets. This opens many opportunities for collectors and people who want to learn and challenge themselves.

Innovation Claim

The innovation BrickBrain gives is that it functions as an adaptive LEGO design coach rather than just being a traditional inventory system or build recommendation application. Many existing applications primarily focus on tracking collections, displaying premade instructions, or recommending existing models. BrickBrain cracks this as it generates personalized build challenges based on a user's inventory, interests, and skill level to encourage original creations to help build their skill development while using BrickBrain.

BrickBrain promotes creativity by giving users the options to choose their difficulty as well as themes and design requirements. By putting both inventory analysis with adaptive challenge generation together, this helps BrickBrain create a learning-focused experience that helps users develop building skills while encouraging experimentation and innovation all together in one place.

Usage Scenario

A user opens up BrickBrain and proceeds to start importing their past sets and some pieces they have laying around. The system then stores and organizes the inventory, allowing the user to select their preferred themes, interests, and challenge difficulty levels to give the system a preference of the user.

BrickBrain analyzes the pieces and breaks down the sets that were imported which then the system would generate a personalized build challenge for the user. For example, a user may receive a challenge to create a city themed vehicle using fewer than 125 pieces, including only 3 wheels. The challenge is specifically designed around the user's inventory and selected preferences.

As the users complete new challenges, they can keep track of their progress and keep receiving new challenges push to encourage users on new and unique building techniques as well as helping with creating thinking. BrickBrain serves as both an intelligent inventory management tool and a personalized LEGO creativity coach that promotes original design and skill development.

SIP BRIEF PART 2

Evaluation Criteria

1. Can users successfully access and navigate the BrickBrain platform?
2. Can users manually add LEGO pieces to their inventory?
3. Can users browse and manage their LEGO inventory?
4. Can users edit or remove inventory items when necessary?
5. Can users select a preferred building theme?
6. Can users select a challenge difficulty level?
7. Can the system generate personalized building challenges?
8. Do generated challenges include clear objectives and requirements?
9. Can users view current and previous challenges?
10. Can users track completed challenges and progress?
11. Does the challenge generation system encourage original LEGO creations?
12. Is the user interface easy to understand and navigate?
13. Does the system function without major errors during testing?
14. Are challenge requirements generated based on user preferences and inventory information?
15. Does the completed prototype demonstrate the concept of an adaptive LEGO design coach?

Goals and Tasks Associated with the Project

Objective 1: Developing the Inventory Management System

The first objective is to create an inventory system where users store, organize, and manage their LEGO collections. This inventory system will be the foundation for challenge generation.

Tasks:

- Research inventory storage methods for LEGO collections.
- Design the database structure used to store LEGO pieces and inventory information.
- Create inventory browsing functionality.
- Develop inventory addition, editing, and deletion features.
- Test inventory storage and retrieval functionality.
- Complete implementation by September 2026.

Objective 2: Develop the Adaptive Challenge Generation System

The second objective is to create a challenge generation system that produces personalized LEGO building challenges based on inventory, interests, and difficulty selections.

Tasks:

- Research challenge-based learning and creativity systems.
- Design challenge generation logic.
- Create theme selection functionality.
- Create difficulty selection functionality.
- Develop challenge requirements, categories, and rules.
- Test generated challenges for usability and creativity.
- Complete implementation by October 2026.

Objective 3: Develop the User Interface

The third objective is to create an active and user-friendly interface that allows users to interact with inventory and challenge systems simple and smoothly.

Tasks:

- Design wireframes and layouts for pages on the application.
- Create inventory management screens.
- Design challenge display screens.
- Develop navigation and user interaction features.
- Have testing done and gather feedback.
- Implement interface improvements based on testing results.

Objective 4: Test and Evaluate the Prototype

The last objective is to make sure that all the major components function correctly and satisfy project goals.

Tasks:

- Test inventory management functionality.
- Test challenge generation functionality.
- Conduct user testing sessions.
- Document bugs and software issues.
- Implement corrections and improvements.
- Evaluate project success using the evaluation criteria established above.

Description of Design Prototype

BrickBrain's design prototype will be developed as a web-based application using HTML, CSS, JavaScript, and database technologies. The prototype will show the core functionality of the BrickBrain platform while also being focused on inventory management while also having active and adaptive challenge generation.

Users will be able to add LEGO pieces and sets to their personal inventory system. The inventory management features will allow users to browse, organize, and update their collections. Users will also be able to select building themes and challenge difficulty levels.

Using inventory information and user preferences, BrickBrain will generate personalized LEGO building challenges. These challenges will contain specific requirements and objectives designed to push creativity, problem-solving, and original design creation rather than replication of existing LEGO models.

The prototype will include a dashboard, inventory browser, inventory import system, challenge generation system, and challenge display screens. The purpose of the design prototype is to demonstrate how BrickBrain functions as an adaptive LEGO design coach that encourages skill development and creative building through personalized challenges.

SIP Brief Part 3 – to be filled out in SIP409

Evaluation Plan

Provide a complete, paragraph-style description of the plan used to evaluate your project. This section should be a **description of the full plan** for how you will go about answering the “Evaluation Criteria” questions. Do not simply repeat the questions!

Note: This section must be revised in SIP409 to describe the full evaluation plan as it was actually implemented.

Project Completion Assessment

Note: This section must be completed in SIP409

Provide an in-depth description of the completion assessment of your project. Describe how well the completed components function and highlight the innovative facets of your design. This is sometimes known as a “Post-Mortem” or “Lessons-Learned Report”. A good approach for this section is to answer the following 4 questions: “What went right? What went wrong? What was learned throughout the process? What would be done differently if you had to do it again?”

Appendices

Note: This section must be completed in SIP408.

Include as appendices any supporting material for this project, including charts, graphs, and other data; images associated with the project; or other documentation (e.g., a game design document or read-me file). Include any prior art that was used such as U.S. Patent Documents, Foreign Patent Documents, or other sources. Remember that this section should only be a list of additional files, not the actual data of the files!

Use the following format:

Appendix letter: description of item – file name

Example...

Appendix A: Game design document – myGameDoc.docx

Appendix B: 3D render of primary character – mainCharacter.jpg